2020 Summer Qual

III.

(a) $\mathbb{E}[y^T W y] = \mathbb{E}[tr(y^T W y)]$

$= \mathbb{E}[tr(W y y^T)]$

$= tr(\mathbb{E}[W y y^T])$

$= tr(W(\Sigma + \mu\mu^T))$

$= \sigma^2 tr(W) + tr(\mu^T W \mu)$

$= \sigma^2 tr(W) + \beta^T X^T W X \beta$

(b) $C\hat{\beta} = C(X^TX)^{-1}X^Ty$: normal

$\mathbb{E}[C\hat{\beta}] = C\beta$

$var(C\hat{\beta}) = \sigma^2 C(X^TX)^{-1}C^T$

$\therefore C\hat{\beta} \sim N(C\beta, \sigma^2 C(X^TX)^{-1}C^T)$

(c) $\hat{\beta}_c = (I-B)\hat{\beta}$

$cov(\hat{\beta}_c, C\hat{\beta}) = (I-B)(X^TX)^{-1}C^T = 0$

(d) $\sigma^2(I-B)(X^TX)^{-1}(I-B) = \sigma^2 \underbrace{(I-B)(X^TX)^{-1}}_{Q}$

Let $y \neq 0, c \neq 0, d = c^T y$

$Qd = 0$ but $d \neq 0$ $\therefore$ rank-deficit

(e) $(C\hat{\beta})^T [C(X^TX)^{-1}C^T]^{-1} (C\hat{\beta})$

$$= y^T \underbrace{X(X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}C(X^TX)^{-1}X^T}_{=A} y$$

$A^2 = A$, $\text{rank}(A) = \text{tr}(A) = q$

$$\frac{(C\hat{\beta})^T [C(X^TX)^{-1}C^T]^{-1} (C\hat{\beta})}{\sigma^2} \sim \chi^2(q, \lambda)$$

where $\lambda = \dfrac{(C\beta)^T [C(X^TX)^{-1}C^T]^{-1} (C\beta)}{2\sigma^2}$

$$\frac{SSE}{\sigma^2} = \frac{y^T(I-H_X)y}{\sigma^2} \sim \chi^2(n-p-1)$$

$A(I-H_X) = 0$

$$\Rightarrow F = \frac{(C\hat{\beta})^T [C(X^TX)^{-1}C^T]^{-1} (C\hat{\beta})/q}{SSE/n-p-1} \sim F_{q, n-p-1, \lambda}$$

$\boxtimes$

(f)

$$C\beta \in (C\hat{\beta})^T [C(X^TX)^{-1}C^T]^{-1} (C\hat{\beta}) \le q \cdot s^2 F_{q, n-p-1}(\alpha)$$

, where $s^2 = \dfrac{SSE}{n-p-1}$

$\boxtimes$

$\boxed{2}$.

(a) Let $Q = A^TA(A^TA)^-$, $P = I$

$$QA^TA = A^TA, \quad PA^TA = A^TA$$

$$0 = (QA^TA - PA^TA)(Q^T - P^T)$$

$$= (QA^T - PA^T)(AQ^T - AP^T)$$

$$= B^TB \Rightarrow B = 0$$

$$AQ^T = AP^T$$

$$\Rightarrow A = A(A^TA)^-A^TA$$

(b) $AA^-A = A \Rightarrow A^T(A^-)^TA^T = A^T \Rightarrow (A^-)^T = (A^T)^-$

$$\Rightarrow (A(A^TA)^-A^T)^T = A((A^TA)^-)^TA^T$$

$$= A(A^TA)^-A^T \quad ; \text{symmetric}$$

$$\underbrace{A(A^TA)^-A^TA}_{=A}(A^TA)^-A^T = A(A^TA)^-A^T \quad ; \text{idempotent}$$

Let G_1 and G_2 s.t. $AG_1A^TA = A$,
$AG_2A^TA = A$

$$0 = (AG_1A^TA - AG_2A^TA)(G_1^TA^T - G_2^TA^T)$$

$$= (AG_1A^T - AG_2A^T)(AG_1A^T - AG_2A^T)^T$$

$$= B^TB \Rightarrow B = 0 \Rightarrow AG_1A^T = AG_2A^T \text{; invariant}$$

$\blacksquare$